

Insurance AI Assistant Fine-Tuning Project

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1. Project Title

Insurance AI Assistant Using QLoRA, Instruction Fine-Tuning, and DPO Alignment.

2. Domain Selected

Commercial Insurance & Underwriting

This project focuses on building an AI-powered insurance assistant capable of answering underwriting, risk assessment, policy evaluation, and insurance-related business questions.

The objective was to adapt a foundation Large Language Model (LLM) to understand insurance-specific terminology, underwriting guidelines, risk management concepts, and respond professionally to users.

3. Business Problem

Insurance professionals spend significant time reviewing submissions, analyzing risks, interpreting policy requirements, and answering repetitive underwriting questions.

Challenges include:

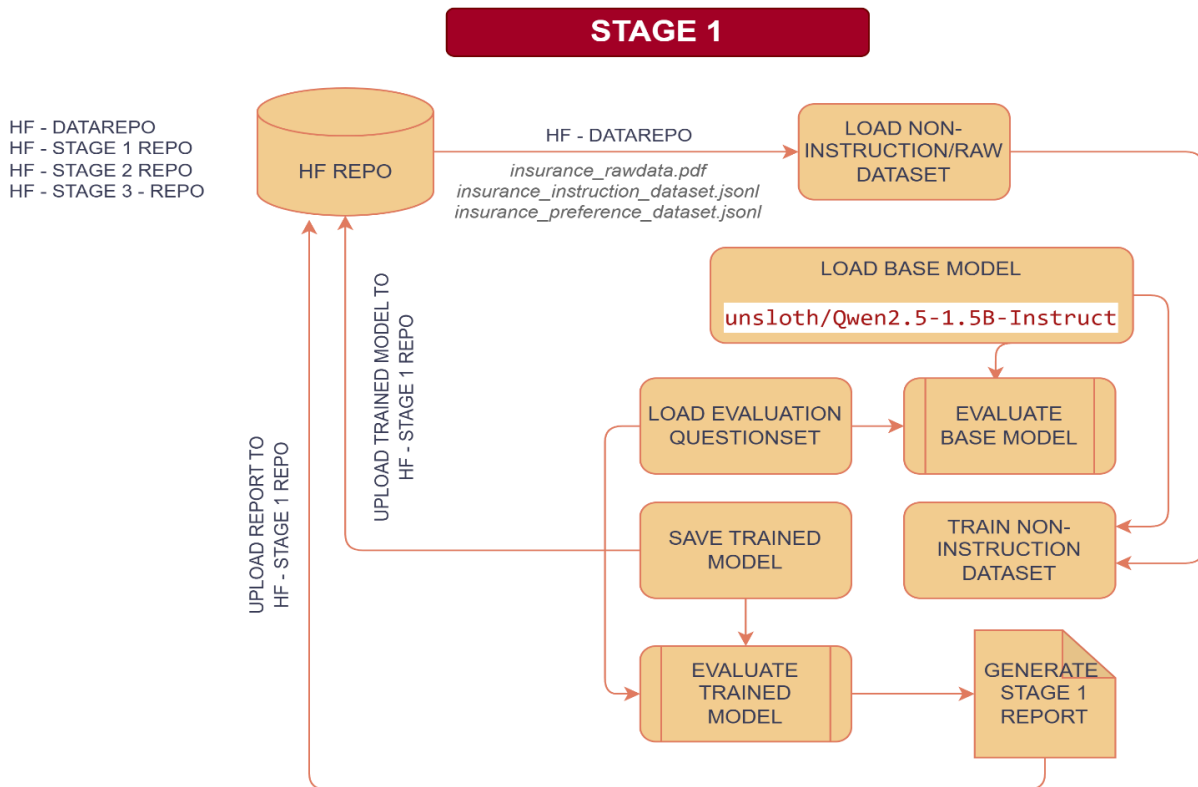
- Manual review of insurance submissions
- Inconsistent underwriting responses
- Slow turnaround time
- Knowledge dependency on experienced underwriters
- Limited accessibility to insurance expertise

The goal of this project is to develop an Insurance AI Assistant capable of:

- Understanding insurance terminology
- Providing underwriting guidance
- Assisting with risk evaluation
- Answering insurance-related questions
- Producing professional and consistent responses

4. Architecture

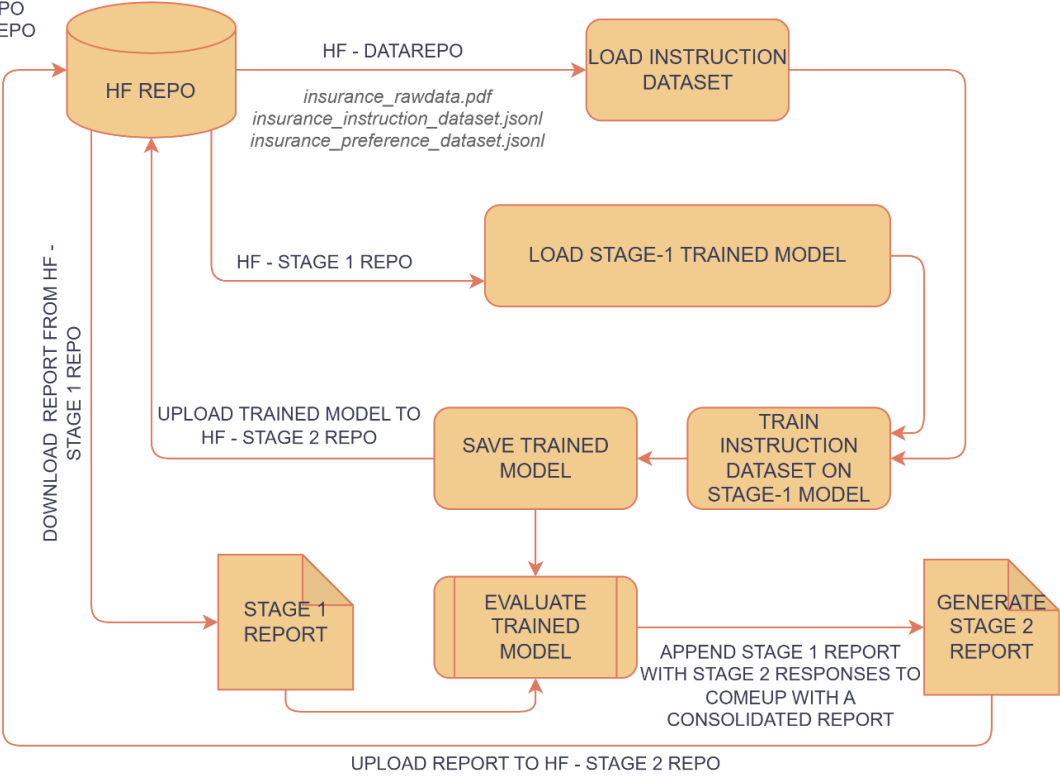
INSURANCE - AI - ASSIST



INSURANCE - AI - ASSIST

STAGE 2

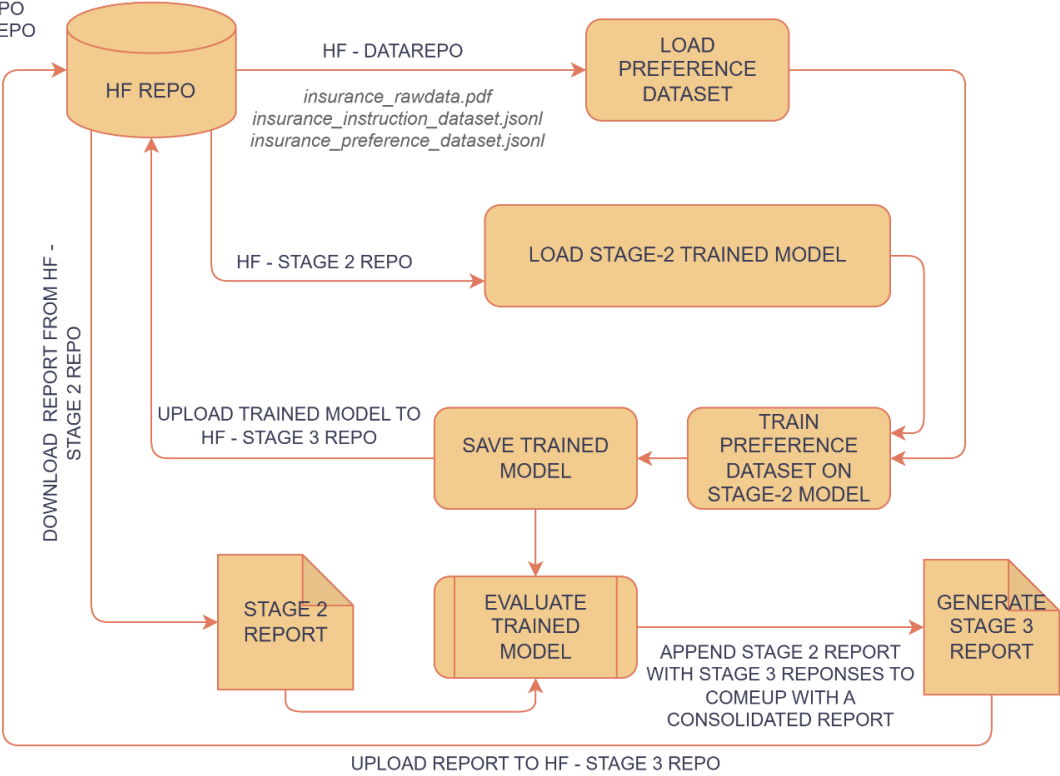
HF - DATAREPO
HF - STAGE 1 REPO
HF - STAGE 2 REPO
HF - STAGE 3 - REPO



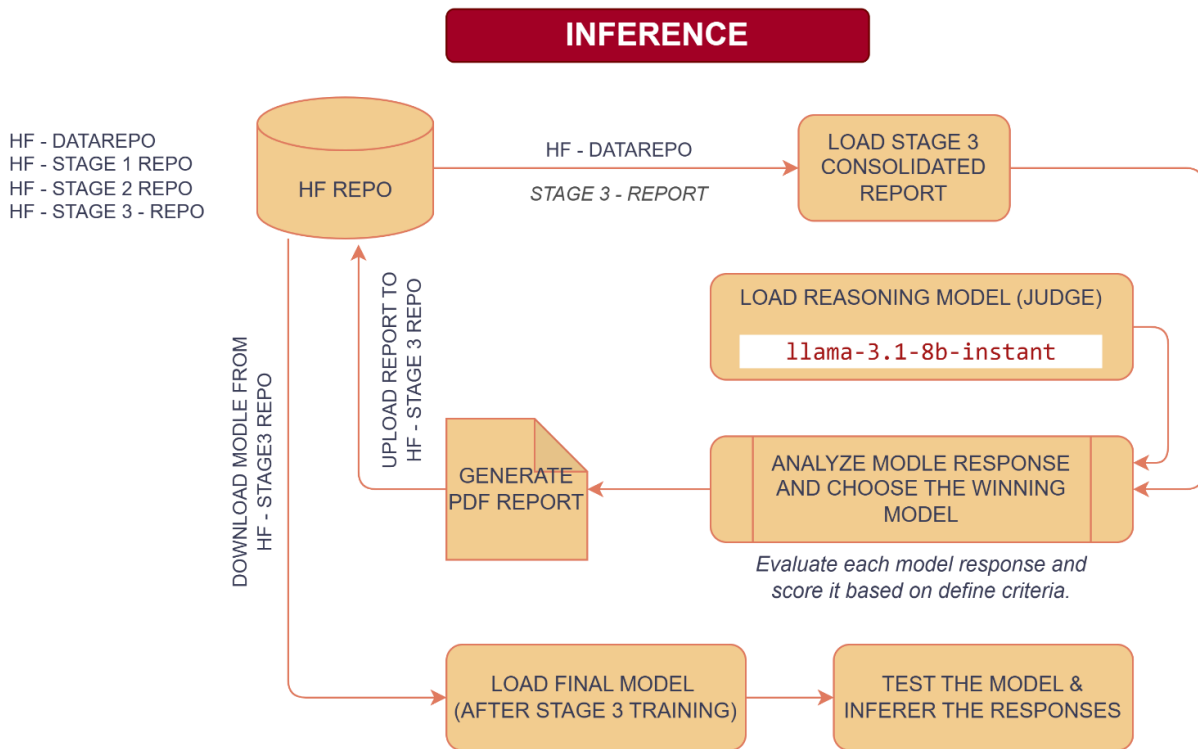
INSURANCE - AI - ASSIST

STAGE 3

HF - DATAREPO
HF - STAGE 1 REPO
HF - STAGE 2 REPO
HF - STAGE 3 - REPO



INSURANCE - AI - ASSIST



5. Dataset Details

Dataset Type

Custom Commercial Insurance Dataset (PDF & JSNOL)

Data Sources

The dataset was created using domain-specific insurance and underwriting content covering:

- Commercial insurance
- Risk assessment
- Workers compensation
- Property insurance
- Liability coverage
- Middle market underwriting
- Submission requirements
- Risk mitigation practices

- Insurance operations

Dataset Transformation Stages

Stage 1 Dataset – Non-instruction insurance knowledge dataset.

Format:

- Question
- Answer

Stage 2 Dataset – Instruction-following dataset.

Format:

- Instruction
- Input
- Response

Stage 3 Dataset – Preference dataset for DPO training.

Format:

- Prompt
- Chosen Response
- Rejected Response

6. Model Used

Base Model – Qwen2.5-1.5B-Instruct

The model was selected because:

- Lightweight and efficient for fine-tuning
- Strong instruction-following capabilities
- Suitable for QLoRA-based training
- Works effectively within Google Colab GPU limitations
- Provides a good balance between performance and computational cost

Judge Model – Llama-3.1-8B-Instant

The judge model was used during the evaluation phase to compare responses from:

- Base Model

- Non-Instruction Fine-Tuned Model
- Instruction Fine-Tuned Model
- DPO Fine-Tuned Model

The judge model assigned scores and identified the best response based on:

- Correctness
- Completeness
- Domain Accuracy
- Professional Quality
- Helpfulness

7. Non-Instruction Fine-Tuning Approach

Objective

Teach the model insurance domain knowledge.

Method

The model was trained on insurance-related question and answer pairs without explicit instruction formatting.

Outcome

The model learned:

- Insurance terminology
- Underwriting concepts
- Risk assessment language
- Commercial insurance knowledge

However, response structure and instruction-following capability remained limited.

8. Instruction Fine-Tuning Approach

Objective

Improve the model's ability to follow user instructions and provide structured responses.

Method

Training examples were converted into instruction-response format.

Outcome

The model became better at:

- Understanding user intent
- Providing structured answers
- Maintaining conversational format
- Producing professional responses

9. DPO Alignment Approach

DPO

Direct Preference Optimization

Objective

Improve answer quality by learning user preferences.

Approach

The model learns which response is preferred and adjusts its behavior accordingly.

Outcome

The model generated:

- More helpful answers
- Better clarity
- Better professionalism
- Reduced hallucinations
- Improved alignment with expected insurance responses

10. LoRA / QLoRA Configuration

QLoRA was selected because it enables efficient fine-tuning on limited GPU resources by loading the model in 4-bit precision while training LoRA adapters.

11. Training Screenshots or Logs

Refer below for training logs –

<https://github.com/Pzazz55/insurance-ai-assistant-finetuning/tree/main/reports>

12. Before vs After Output Comparison

Refer the below final evaluation report –

https://github.com/Pzazz55/insurance-ai-assistant-finetuning/blob/main/reports/final_evaluation_20260710_025732.pdf

13. Final Observations

Key Findings

- Domain adaptation significantly improved insurance knowledge.
- Instruction tuning improved response structure.
- DPO alignment improved response quality and professionalism.
- QLoRA enabled efficient training on limited hardware.
- Preference optimization produced the most consistent and useful outputs.

14. Challenges Faced

Technical Challenges

- Limited GPU memory in Google Colab
- Token length limitations
- Parsing large evaluation reports
- Managing multiple training stages
- DPO dataset creation and validation
- Consistent evaluation across model checkpoints

Solutions

- Used QLoRA for memory efficiency
- Applied truncation strategies
- Built automated evaluation pipelines
- Implemented model comparison frameworks

15. Future Improvements

Planned Enhancements

1. Retrieval Augmented Generation (RAG)

Integrate:

- Underwriting manuals
- Policy documents
- Regulatory references

2. Insurance Knowledge Base

Create a searchable repository of insurance documents.

3. Vector Database Integration

Implement a Vector Database like – FAISS, ChromaDB, Pinecone, etc.

4. Multi-Agent Architecture

Develop specialized agents:

- Underwriting Agent
- Claims Agent
- Risk Assessment Agent
- Compliance Agent

5. Web Application

Convert the solution into an interactive application using:

6. Continuous Preference Learning

Expand DPO datasets using production feedback and user interactions.

16. Important Links

- GitHub – <https://github.com/Pzazz55/insurance-ai-assistant-finetuning/tree/main>
- HuggingFace – <https://huggingface.co/Pzazz55>

Conclusion

This project successfully developed a domain-adapted Insurance AI Assistant through a three-stage fine-tuning pipeline consisting of:

1. Non-Instruction Fine-Tuning

2. Instruction Fine-Tuning
3. DPO Preference Alignment

Using QLoRA enabled efficient training on limited GPU resources while achieving strong performance in commercial insurance and underwriting tasks. The final DPO-aligned model demonstrated the highest quality, professionalism, and user alignment among all evaluated model versions.

